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IARCO RESEARCH PROPOSAL

SIEVE

Signature-Informed Evaluation of Variant Escape

A Within-Patient Counterfactual Test of Whether HLA-Restricted Immune Selection on the Cancer Genome Is Carcinogen-Specific

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Research Topic: Cancer Immunogenomics / Computational Biology
Title: SIEVE: A Within-Patient Counterfactual Test of Whether HLA-Restricted Immune Selection on the Cancer Genome Is Carcinogen-Specific

Research Problem

Over the past two decades, Immune checkpoint inhibitors have led to amazing healing in patients who would've had months to live, making it an extremely viable form of cancer therapy. Unfortunately, they also fail most of the people who receive them. Across tumours, objective response rates to single-agent PD-1 blockade are around fifteen to thirty percent, and the treatments lead to serious immune-related toxicity and extremely high annual costs for treatment. The real problem is that there is no solid way to predict who they will work for, and the money and side effects have to be endured regardless of whether the treatment works or not. Tumour mutations and PD-L1 expression are the biomarkers commonly used to find issues, and both have their own flaws: each explains only a small amount of the variance in each patient's response, and neither looks at the patient's genome, adding basically no personality to treatment.

One solution looks not at the tumour but at the actual patient. Every nucleated cell has fragments of its internal proteins on its surface using MHC class I molecules, encoded by HLA-A, HLA-B and HLA-C genes. When a cell mutates, T cells can recognise it only if HLA molecules can bind and flag the mutated peptide to the T cell. If a patient's HLA cannot highlight a mutated cell, it goes unnoticed by their immune system, essentially going invisible. A tumour takes years to fully develop, and by that time, mutations that a person's HLA can clearly highlight are thoroughly removed by the immune system. Therefore, a cancerous tumour has a high concentration of mutations that HLA molecules cannot identify and remove [1]. According to this finding, a person's HLA genotype determines which cancers they can develop by determining which mutations it can successfully bind to and aid in immune response.

Researchers have tested this, but no conclusion has been reached. The original analysis reported the effect and quantified it with a per-patient, per-mutation presentation score, testing against every single form of cancer [1]. A later reanalysis argued that the association does not account for statistical non-independence among those scores, and found that randomizing patients' HLA genotypes reproduced the reported signal, disproving the initial finding [3]. 10 years later, one of the central questions in cancer immunogenetics - whether the immune system is the reason behind which mutations form in cancerous tumors - has not been solved.

This proposal identifies a reason the two results could both be correct, and outlines a test designed to both apart. If immune selection is **carcinogen-specific**, then a pooled (pan-cancer) analysis that mixes tumour types with different dominant processes would mute a real effect toward 0. That possibility has not been tested at the patient level, which is what I propose to research.

Existing Literature

Cancer immunoediting is a process where the immune system gets rid of and modifies growing tumors to protect the body, and is a long-standing system[4]. Its effect on a patient's genome is hard to address because the goal is to find cancerous mutations that haven't been found by modern medicine, essentially looking for a needle in a haystack. Marty et al. addressed this by constructing the Patient Harmonic-mean Best Rank (PHBR) score, which shows how well a given patient's complete class I genotype highlights peptides arising from mutations, and reported that, across roughly 1,000 recurrent oncogenic mutations, those mutations tended to occur in patients who present them poorly [1]. They also extended the argument to MHC class II [2].

Clinical testing has also proved the significance of HLA. Having a homozygous HLA genotype is found to cause worse survival under checkpoint blockade [5], and greater HLA evolution is associated with better immune system outcomes [6]. Germline variation has also been proven to shape the tumour immune microenvironment [7].

Kherreh, Cleary and Seoighe argued against this claim in their research paper [3]. They claim that because PHBR is a mutation-allele pair property, a substitution leads to many observations across every patient sharing the relevant alleles that are correlated. So, for that reason, treating those observations as independent inflates the significance of the finding. After accounting for this structure, the reported effect was not distinguishable from what randomised genotypes produced. The critique is statistical, and it has not been rebutted with a design that avoids the problem rather than adjusting for it.

A third result reframes the question. A 2026 amino acid analysis shows that different mutational processes generate systematically different peptide chemistry. They found different results for different signatures. For example, the AAS4 signature yields poorly immunogenic neopeptides, but the AAS5 signature is enriched for immunogenic ones, and HLA-B*07:02 highlights proline-rich mutations very well, essentially counteracting the initial pattern [8]. The authors state that patient-level HLA analysis fell outside that study's scope, making it an up-and-coming topic to research. This altogether suggests the pooled question may simply be the wrong question, and that further research is necessary.

A similar problem has a solution in a different field of medicine. In vaccine efficacy trials, **sieve analysis** is a process where you find whether protection varies with the genotype of the breakthrough pathogen, treating the vaccine as a sieve that blocks some variants and passes others [9]; it was applied to the RV144 HIV-1 trial to show that efficacy differed by Env V2 genotype [10]. The design proposed here is a sieve analysis of the cancer genome, in which the sieve is the patient's HLA genotype and the variant classes are the mutations.

Research Question

Primary question. Is HLA immune selection on the cancer genome carcinogen-specific? So, does the strength of selection depend on which mutational process produced the mutation?

I broke this primary question down into four sub-questions:

1. Do the COSMIC substitution signatures create different amino acid substitution spectra, and does predicted MHC class I presentation change across HLA alleles as a function of those spectra?
2. Within a tumour, is the mutation worse by that patient's own HLA genotype than the counterfactual substitutions that the tumour's dominant signature could have made at the same site?
3. Does the magnitude of the within-patient effect change between tumours with different mutational processes: ex. ultraviolet light in melanoma, tobacco in lung adenocarcinoma, APOBEC in bladder carcinoma?
4. Does the effect become invisible in synonymous mutations and in passenger mutations outside known cancer genes, where immune selection shouldn’t be working?

Methodology

The study is computational and uses exclusively public, de-identified data. I organized it into 5 phases.

Phase 1: Construction of the SIEVE atlas. For each COSMIC signature (version 3.4), the trinucleotide-context substitutions will be propagated through the entirety of the human coding sequence in order to reach the amino acid substitutions that may be anticipated. The predicted peptide binding affinity evaluations will then be made for the roughly 100 most frequently encountered HLA-A, -B and -C alleles by means of NetMHCpan-4.1 [11], with results aggregated into PHBR-style per allele presentability values. Therefore, the outcome will be a signature x allele presentability matrix. Since this output has not been found in the scientific publications, it will be published in a publicly available format with a reference to our study.

Phase 2: Cohort and genotype assembly. Collection of cohorts and genotypes. Somatic mutations will be derived from the TCGA MC3 open-access database [12]. Class I HLA genotypes for individual patients will be taken from the supplementary tables of the published checkpoint blockade studies [13], [14], with POLYSOLVER [15] and arcasHLA [16] as backup options if sequencing permits. The analysis will start with three types of cancer distinguished by the pattern of mutations: skin melanoma (UV), lung adenocarcinoma (tobacco), and bladder carcinoma (APOBEC).

Phase 3: The within-patient counterfactual test. For each non-synonymous mutation observed, the alternative substitutions possible in that codon based on the tumor's signature are computed and have their signature probabilities applied to weight them. The known mutation and its counterfactuals are compared to the actual genotype of the patient, with the SIEVE score being the rank of the observed mutation within its counterfactual population. Assuming that selection is not playing a part in the substitution at that locus, the alternative mutation ranks are distributed uniformly around their means, which come out to be (k+1)/2, where k is the number of alternative mutations. In this way, the null becomes exact because no distributional assumptions are made about the comparison, and it is restricted to one patient at one locus only.

Phase 4: Negative controls and null sensitivity. The same pipeline will be repeated on synonymous mutations and on silent mutations in non-cancerous genes. Three counterfactual null models will be produced, and all of them will be reported: signature-matched, trinucleotide-matched, and uniform models. The sensitivity of the resulting methods across counterfactual models will be isolated and reported. Predicted binding will be grounded on experimentally verified epitopes taken from IEDB [17] and CEDAR [18], and the whole procedure will be repeated with MHCflurry 2.0 [19] as a second independent predictor.

Phase 5: Power analysis and pre-registration. In order to define a null result, power analysis will be done prior to the experiment, in order for the theoretically possible observation of effect sizes to be reflected in the intended test. Additionally, the stopping rules for the data collection process, as well as the null models and analysis code, will be submitted to open storage before any data is looked at.

Preliminary testing of the design. We tested the counterfactual-rank prototype using synthetic data with known ground truth. Across 100 simulations per condition, it produced 2 significant results without injected selection and detected the injected signal in all 100 simulations at the chosen selection parameter of 0.30. Testing used 400 synthetic patients, 100 loci, and 10 abstract candidate substitutions, with a one-sided significance threshold of 0.05. The corresponding exact 95% confidence intervals were approximately 0.24%–7.04% for null rejections and 96.38%–100% for signal detection.

These results concern synthetic presentation scores under a uniform mutation model. They do not establish clinical validity, validate the signature-weighted test or constitute a complete power analysis. Patient-level inference and handling of tied predictions still require validation.

Research Topic

This subject is current, pending, and has a bearing on clinical practice. Currently, checkpoint blockade represents a treatment available for several cancer forms that still has not provided any results for a significant number of patients receiving it. The inconsistency of results in this field relates to the central unresolved issue: how patients should be classified to achieve success in the use of checkpoint blockade. If HLA-based pretreatment selection is truly specific to carcinogen, this has two important implications. First, the dispute that has existed for the last ten years is resolved since neither researcher has actually made any mistakes, as it has been confirmed that the results can be expected in heterogeneous cancers. The second, and perhaps more important, implication is that the level of protection offered by HLA genotype from different exposures is different as well; thus, any potential future developments in germline HLA use for risk assessment and immunotherapy should account for the process of mutation that triggered cancer development, while global studies that ignore the divergence of processes will be ineffective in producing the desired results. The proposal is also unusual in what it declines to build: it deliberately avoids adding another machine-learning predictor to a saturated area, and instead imports a validated statistical design from vaccine trial methodology into a field that has not used it.

Expected Outcomes

Three outcomes are anticipated, and all three are informative. If the effect holds within patients across all three tumour types, a contested result in cancer immunology will have been recovered using a design immune to the objection that defeated it. If the effect is absent under a clean design and adequate power, that is a bounded negative finding about a framework still being built upon, stated as an excluded effect size rather than as a failure to find one. If the effect holds for some mutational processes and not others, immune selection is carcinogen-specific — the most interesting of the three, and the outcome that reconciles the existing disagreement. In every branch, the SIEVE Atlas produced in Phase 1 is released as a standalone public resource.

Limitations

Predicted HLA binding does not guarantee peptide presentation, T-cell recognition or tumour-cell destruction. The counterfactual probabilities depend on estimated mutational signatures, and errors in those estimates could produce an apparent selection signal. Functional selection and mutation-detection biases could also affect the results. Matched mutation and HLA

data may limit sample size, while repeated observations within patients and at recurrent loci require appropriate statistical treatment. Comparisons between cancer types may reflect tissue differences as well as mutational processes. Finally, the completed simulations assess a simplified setting and do not establish performance on real patient data.

Project Timeline

The project will last about eight months. The first two months will be spent on the construction and validation of the SIEVE Atlas. The third through fifth months will concern the assembly of the cohort, testing within-patient counterfactuals, and use of the complete negative control suite. The sixth through seventh months will involve investigating the HLA loss of heterozygosity as a natural experiment in tumours defined by mutations that occurred post-loss [20]. The eighth month will be designated for writing, depositing code and data, and making the results ready for presentations.

Post-Program Plan

The SIEVE Atlas is going to be archived as a permanent dataset, categorized by version and having a permanent ID in its classification. This work will be presented to a peer-reviewed publication in the area of computational biology/cancer biology. A presentation at undergraduate research conferences will be prepared as well. Relevant laboratories researching HLA-restricted antigen presentation will be contacted to point out any mistakes before the publication takes place.

Quality of Writing

This proposal is not only directed towards experts in cancer immunogenomics but rather meant for a broader audience. The technical terminology is made clear with proper definitions. The hypothesis is put forward in a simple way with no excessive formality. All claims made in the literature are followed with IEEE-style citations. The structure adheres to the standard format for research proposals, ensuring compliance with IARCO requirements.

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